

Questions & Answers

Remote Sensing, GIS & Image Processing — 349 Questions

1. Digital image processing primarily deals with:

Answer: Numerical representation of images

2. A digital image is composed of:

Answer: Pixels

3. Pixel intensity values represent:

Answer: Brightness

4. Image resolution refers to:

Answer: Pixel density

5. The smallest element of an image is:

Answer: Pixel

6. Radiometric resolution indicates:

Answer: Gray level sensitivity

7. Spatial resolution is determined by:

Answer: Pixel size

8. Spectral resolution depends on:

Answer: Number of spectral bands

9. Temporal resolution refers to:

Answer: Time between acquisitions

10. Image enhancement aims to:

Answer: Improve visual quality

11. Histogram represents:

Answer: Frequency of gray levels

12. Contrast stretching improves:

Answer: Visual distinction

13. Noise in an image is:

Answer: Random variation

14. Image filtering is used to:

Answer: Remove unwanted variations

15. A low-pass filter is used for:

Answer: Smoothing

16. A high-pass filter is mainly used for:

Answer: Edge enhancement

17. Image segmentation refers to:

Answer: Dividing image into regions

18. Thresholding is a:

Answer: Segmentation technique

19. Binary images contain:

Answer: Two intensity levels

20. Image compression aims to:

Answer: Reduce storage

21. Lossy compression:

Answer: Removes some information

22. Lossless compression:

Answer: Preserves original data

23. JPEG is a:

Answer: Lossy compression format

24. PNG is mainly:

Answer: Lossless

25. RGB model consists of:

Answer: Red, Green, Blue

26. Grayscale images use:

Answer: Single intensity channel

27. Sampling in images refers to:

Answer: Spatial measurement

28. Quantization deals with:

Answer: Intensity levels

29. Image restoration focuses on:

Answer: Recovering degraded images

30. Edge detection identifies:

Answer: Object boundaries

31. Sobel operator is used for:

Answer: Edge detection

32. Morphological operations work on:

Answer: Shape and structure

33. Dilation operation:

Answer: Expands objects

34. Erosion operation:

Answer: Shrinks objects

35. Opening operation is erosion followed by:

Answer: Dilation

36. Closing operation is dilation followed by:

Answer: Erosion

37. Image interpolation is used during:

Answer: Resampling

38. Nearest neighbor interpolation is:

Answer: Fastest method

39. Image mosaicking combines:

Answer: Multiple images

40. Image registration aligns images based on:

Answer: Common features

41. Raw image data is also known as:

Answer: Level-0 data

42. Level-1 data includes:

Answer: Radiometric correction

43. Level-2 data represents:

Answer: Extracted information

44. Primary data products come directly from:

Answer: Satellite sensors

45. Secondary data products are:

Answer: Derived from primary data

46. Image map products include:

Answer: Topographic maps

47. DEM stands for:

Answer: Digital Elevation Model

48. Orthophotos are:

Answer: Geometrically corrected images

49. Thematic maps show:

Answer: Specific features

50. Land use maps represent:

Answer: Human activity patterns

51. False color composites use:

Answer: Non-natural band combinations

52. True color images resemble:

Answer: Human vision

53. Multispectral data contains:

Answer: Multiple spectral bands

54. Hyperspectral data contains:

Answer: Hundreds of bands

55. Panchromatic images are:

Answer: Black and white

56. Data fusion combines:

Answer: Multiple datasets

57. GIS-ready products are:

Answer: Geo-referenced

58. Image tiling is used to:

Answer: Manage large datasets

59. Metadata provides:

Answer: Image information

60. Geo-referencing assigns:

Answer: Geographic coordinates

61. Vector data represents features as:

Answer: Points, lines, polygons

62. Raster data represents features as:

Answer: Pixels

63. Continuous data example:

Answer: Elevation

64. Discrete data example:

Answer: Land cover

65. Image classification produces:

Answer: Labeled images

66. Supervised classification requires:

Answer: Training samples

67. Unsupervised classification groups pixels based on:

Answer: Similarity

68. Accuracy assessment measures:

Answer: Classification correctness

69. Confusion matrix evaluates:

Answer: Classification accuracy

70. Kappa coefficient measures:

Answer: Agreement level

71. Image interpretation involves:

Answer: Visual analysis

72. Image interpretation is mainly:

Answer: Human-based

73. Interpretation strategy depends on:

Answer: Objective of study

74. Interpretation keys help in:

Answer: Feature identification

75. Tone refers to:

Answer: Brightness variation

76. Color interpretation depends on:

Answer: Band combination

77. Shape refers to:

Answer: Outline of objects

78. Size of an object is determined by:

Answer: Relative scale

79. Pattern indicates:

Answer: Spatial arrangement

80. Texture describes:

Answer: Smoothness or roughness

81. Association refers to:

Answer: Relationship with surroundings

82. Shadow helps in identifying:

Answer: Height and shape

83. Site refers to:

Answer: Physical setting

84. Context interpretation uses:

Answer: Surrounding features

85. Direct interpretation uses:

Answer: Visual observation

86. Indirect interpretation uses:

Answer: Inference

87. Visual interpretation is best for:

Answer: Complex patterns

88. Keys are developed from:

Answer: Experience and reference data

89. Interpretation keys vary with:

Answer: Scale and resolution

90. High resolution images improve:

Answer: Feature recognition

91. Multiscale interpretation involves:

Answer: Multiple resolutions

92. Visual fatigue is reduced by:

Answer: Proper interpretation tools

93. Interpretation accuracy improves with:

Answer: Experience

94. Ground truth data supports:

Answer: Validation

95. Stereo interpretation provides:

Answer: Height perception

96. Temporal comparison helps identify:

Answer: Changes over time

97. Manual interpretation is:

Answer: Subjective

98. Automated interpretation relies on:

Answer: Algorithms

99. Hybrid interpretation combines:

Answer: Manual and automated methods

100. Interpretation confidence increases with:

Answer: Multiple evidences

101. Repetition in patterns indicates:

Answer: Cultural features

102. Linear patterns often indicate:

Answer: Roads

103. Curvilinear patterns often represent:

Answer: Rivers

104. Rectangular patterns suggest:

Answer: Human activities

105. Interpretation keys must be:

Answer: Flexible

106. Visual interpretation is limited by:

Answer: Human perception

107. Interpretation errors arise due to:

Answer: Misidentification

108. Seasonal variation affects:

Answer: Appearance of features

109. Interpretation reliability increases with:

Answer: Multi-source data

110. Interpretation keys are most useful in:

Answer: Visual analysis

111. Topography refers to:

Answer: Surface features of Earth

112. Contours represent:

Answer: Equal elevation

113. Closely spaced contours indicate:

Answer: Steep slope

114. Widely spaced contours indicate:

Answer: Flat terrain

115. Hill appears as:

Answer: Closed contours

116. Valley is represented by:

Answer: V-shaped contours

117. Ridge appears as:

Answer: V pointing uphill

118. Drainage refers to:

Answer: Flow of water

119. Dendritic drainage resembles:

Answer: Tree branches

120. Trellis drainage occurs in:

Answer: Folded terrain

121. Radial drainage flows:

Answer: Away from center

122. Centripetal drainage flows:

Answer: Toward center

123. Parallel drainage occurs on:

Answer: Steep slopes

124. Rectangular drainage indicates:

Answer: Faulted terrain

125. Drainage density depends on:

Answer: All of the above

126. High drainage density indicates:

Answer: Impermeable rock

127. Low drainage density indicates:

Answer: Permeable material

128. Texture in image interpretation refers to:

Answer: Surface roughness

129. Coarse texture indicates:

Answer: Rough surface

130. Fine texture indicates:

Answer: Smooth terrain

131. Urban areas usually show:

Answer: Coarse texture

132. Agricultural fields show:

Answer: Regular patterns

133. Erosion refers to:

Answer: Removal of soil

134. Sheet erosion removes:

Answer: Thin soil layer

135. Rill erosion forms:

Answer: Small grooves

136. Gully erosion forms:

Answer: Deep channels

137. Erosion is highest in:

Answer: Steep slopes

138. Vegetation reduces erosion by:

Answer: Holding soil

139. Water erosion is common in:

Answer: High rainfall areas

140. Wind erosion dominates in:

Answer: Deserts

141. Erosion features are identified using:

Answer: Tone and texture

142. Alluvial fans show:

Answer: Fan-shaped deposits

143. Delta forms at:

Answer: River mouth

144. Meandering rivers indicate:

Answer: Flat terrain

145. Braided rivers indicate:

Answer: High sediment load

146. Flood plains show:

Answer: Flat low-lying areas

147. Badland topography shows:

Answer: Highly eroded terrain

148. Karst topography is formed in:

Answer: Limestone

149. Sinkholes indicate:

Answer: Karst terrain

150. Glacial erosion forms:

Answer: U-shaped valleys

151. Cirques are formed by:

Answer: Glaciers

152. Moraines are:

Answer: Glacial deposits

153. Coastal erosion is caused by:

Answer: Waves

154. Cliffs indicate:

Answer: Steep erosion

155. Erosion control measures include:

Answer: Terracing

156. Texture analysis helps identify:

Answer: Land cover types

157. Rough texture often indicates:

Answer: Forests

158. Smooth texture often indicates:

Answer: Water bodies

159. Drainage anomalies indicate:

Answer: Structural control

160. Terrain analysis supports:

Answer: All of the above

161. Stereoscope is used for:

Answer: 3D viewing

162. Pocket stereoscope is:

Answer: Portable

163. Mirror stereoscope uses:

Answer: Mirrors and lenses

164. Stereoscopic vision helps in:

Answer: Height perception

165. Light table is used for:

Answer: Viewing transparencies

166. Magnifying glass helps in:

Answer: Detail enhancement

167. Zoom lens is used to:

Answer: Magnify features

168. Parallax bar measures:

Answer: Height difference

169. Image interpretation software assists in:

Answer: Digital analysis

170. GIS software integrates:

Answer: Spatial and attribute data

171. Digital display quality depends on:

Answer: Screen resolution

172. Monitors used for interpretation should have:

Answer: High color depth

173. Hardcopy prints are limited by:

Answer: Printing quality

174. Analog interpretation uses:

Answer: Photographic prints

175. Digital interpretation allows:

Answer: Zoom and enhancement

176. Stereo pairs require:

Answer: Overlapping images

177. Image overlays help in:

Answer: Feature comparison

178. GPS supports interpretation by providing:

Answer: Location accuracy

179. Field verification is also known as:

Answer: Ground truthing

180. Interpretation reliability increases with:

Answer: Multiple tools

181. Digital stereoscopes use:

Answer: Software

182. Image annotation tools help in:

Answer: Labeling features

183. Virtual globes assist in:

Answer: Visualization

184. Interpretation equipment should be:

Answer: Calibrated

185. Calibration ensures:

Answer: Measurement accuracy

186. Image interpretation is improved by:

Answer: Proper lighting

187. Stereoscopic overlap is typically:

Answer: 60%

188. Viewing angle affects:

Answer: Perception

189. Interpretation workstation includes:

Answer: Hardware and software

190. Digital zoom differs from optical zoom because it:

Answer: Enlarges pixels

191. Stereo effect is based on:

Answer: Parallax

192. Interpretation equipment must be:

Answer: User-friendly

193. Accurate interpretation requires:

Answer: Proper tools and experience

194. Visual fatigue can be reduced by:

Answer: Proper breaks

195. Interpretation errors are reduced by:

Answer: Multiple data sources

196. Visualization tools help in:

Answer: Pattern recognition

197. Stereo interpretation is limited by:

Answer: Overlap quality

198. Interpretation equipment selection depends on:

Answer: Application needs

199. Visual interpretation complements:

Answer: Automated methods

200. Effective image interpretation combines:

Answer: Tools and experience

201. Remote sensing is best defined as:

Answer: Acquisition of information about an object without physical contact

202. Which of the following is the first step in the remote sensing process?

Answer: Energy source or illumination

203. The Sun acts as a primary source of energy in:

Answer: Passive remote sensing

204. Ground truth data is mainly used to:

Answer: Verify and validate remote sensing data

205. Which stage converts raw data into usable image format?

Answer: Pre-processing

206. Remote sensing data is collected using:

Answer: Sensors

207. The platform that carries sensors is known as:

Answer: Satellite

208. Which of the following is NOT a step in the remote sensing process?

Answer: Manual cartography

209. Remote sensing helps mainly in:

Answer: Large-area observation

210. Remote sensing observations are usually:

Answer: Non-contact and repetitive

211. Passive remote sensing depends on:

Answer: Reflected solar radiation

212. Active remote sensing systems include:

Answer: RADAR and LiDAR

213. Which system can operate day and night?

Answer: Active sensors

214. Radar remote sensing operates mainly in:

Answer: Microwave region

215. Passive sensors cannot operate effectively during:

Answer: Nighttime

216. LiDAR primarily uses:

Answer: Laser light

217. Which of the following is an example of passive sensor?

Answer: Thermal infrared sensor

218. Active remote sensing is preferred in:

Answer: Cloudy and night conditions

219. Microwave sensing is advantageous because it:

Answer: Penetrates clouds

220. Active sensors measure:

Answer: Backscattered energy

221. The electromagnetic spectrum is arranged based on:

Answer: Frequency or wavelength

222. Visible spectrum ranges approximately from:

Answer: 0.4–0.7 μm

223. Which band is most suitable for vegetation studies?

Answer: Near Infrared

224. Thermal infrared radiation is associated with:

Answer: Emission

225. Shorter wavelength implies:

Answer: Higher energy

226. Which region is used in radar remote sensing?

Answer: Microwave

227. Atmospheric window refers to:

Answer: Transmission zone

228. Ultraviolet radiation has:

Answer: High frequency

229. The EM spectrum used for night temperature measurement is:

Answer: Thermal IR

230. Which radiation is least affected by weather?

Answer: Microwave

231. Spatial resolution refers to:

Answer: Pixel size

232. Higher spatial resolution means:

Answer: Smaller pixel size

233. Spectral resolution depends on:

Answer: Bandwidth

234. Temporal resolution indicates:

Answer: Revisit time

235. Radiometric resolution represents:

Answer: Brightness levels

236. 8-bit data has radiometric resolution of:

Answer: 256 levels

237. High temporal resolution is important for:

Answer: Change detection

238. Which resolution affects object identification?

Answer: Spatial

239. Multispectral images have:

Answer: Multiple bands

240. Hyperspectral images contain:

Answer: Many narrow bands

241. Scattering occurs mainly due to:

Answer: Atmospheric particles

242. Rayleigh scattering affects:

Answer: Short wavelengths

243. Mie scattering is caused by:

Answer: Large particles like dust

244. Non-selective scattering affects:

Answer: All wavelengths equally

245. Absorption reduces:

Answer: Signal strength

246. Ozone absorbs radiation in:

Answer: Ultraviolet region

247. Water vapor absorbs radiation mainly in:

Answer: Thermal IR

248. Atmospheric windows are regions of:

Answer: High transmission

249. Atmospheric effects are minimum in:

Answer: Microwave region

250. Haze in images is mainly due to:

Answer: Scattering

251. A sensor records:

Answer: Reflected or emitted energy

252. Pushbroom scanners capture data:

Answer: Line by line

253. Whiskbroom scanners use:

Answer: Rotating mirror

254. Geostationary satellites revolve at:

Answer: 36,000 km

255. Polar-orbiting satellites provide:

Answer: High spatial resolution

256. IRS satellites belong to:

Answer: India

257. Landsat is operated by:

Answer: NASA/USGS

258. Sun-synchronous satellites maintain:

Answer: Constant local time

259. Sensor IFOV determines:

Answer: Pixel size

260. Multispectral sensors record data in:

Answer: Several bands

261. A map is a:

Answer: Symbolic representation of Earth

262. Point, line and polygon are:

Answer: Spatial elements

263. Large-scale maps show:

Answer: Small area, more detail

264. RF scale 1:50,000 indicates:

Answer: Reduced map

265. GIS software is mainly used for:

Answer: Spatial data analysis

266. Example of GIS software:

Answer: ArcGIS

267. Coordinate system defines:

Answer: Location on Earth

268. UTM system divides Earth into:

Answer: 60 zones

269. Raster data is represented as:

Answer: Pixels

270. Vector data includes:

Answer: Points, lines, polygons

271. Manual map production is replaced by:

Answer: Computer-aided mapping

272. Satellite classification based on orbit includes:

Answer: Polar & Geostationary

273. Satellite classification based on purpose includes:

Answer: Weather, Communication, Remote sensing

274. Weather satellites mainly monitor:

Answer: Clouds and storms

275. Communication satellites operate mainly in:

Answer: GEO

276. Resource satellites are used for:

Answer: Earth observation

277. GPS satellites belong to:

Answer: Navigation

278. Cartographic output includes:

Answer: Maps and layouts

279. Spatial referencing links:

Answer: Image to ground coordinates

280. Datum refers to:

Answer: Reference surface

281. GIS integrates:

Answer: Data, hardware, software, people

282. Digital maps are advantageous because they are:

Answer: Easy to update

283. Spatial data analysis helps in:

Answer: Decision making

284. Thematic maps show:

Answer: Specific themes like rainfall

285. Topographic maps show:

Answer: Relief and elevation

286. Contour lines represent:

Answer: Elevation

287. Map legend explains:

Answer: Map symbols

288. North direction in maps is usually:

Answer: Top

289. Remote sensing and GIS together support:

Answer: Spatial decision systems

290. Satellite revisit time relates to:

Answer: Temporal resolution

291. False color composite uses:

Answer: Assigned colors

292. Image classification groups pixels based on:

Answer: Spectral similarity

293. Unsupervised classification uses:

Answer: Algorithms to form clusters

294. GIS is widely used in:

Answer: All of the above

295. Satellite images differ from maps because they are:

Answer: Raw data representation

296. Remote sensing reduces:

Answer: Field work

297. Satellite imagery is useful for:

Answer: Regional monitoring

298. Map projection converts:

Answer: Sphere to plane

299. Remote sensing is most effective for:

Answer: Continuous Earth monitoring

300. Remote sensing observations are generally recorded in the form of:

Answer: Digital numbers

301. The term “platform” in remote sensing refers to:

Answer: Satellite or aircraft carrying the sensor

302. Which stage involves converting signals into images?

Answer: Data processing

303. Ground truthing improves:

Answer: Data accuracy

304. Remote sensing is preferred over conventional surveys because it:

Answer: Provides synoptic view

305. Which of the following is NOT a remote sensing platform?

Answer: GPS receiver

306. The output of remote sensing is generally:

Answer: Both qualitative and quantitative

307. Remote sensing data can be:

Answer: Either analog or digital

308. The concept of repetitive coverage is essential for:

Answer: Change detection

309. Remote sensing mainly supports:

Answer: Earth resource monitoring

310. Passive remote sensing cannot operate effectively under:

Answer: Cloudy conditions

311. Active sensors emit their own energy to:

Answer: Illuminate target

312. Synthetic Aperture Radar (SAR) is an example of:

Answer: Active microwave sensor

313. Which remote sensing system is least affected by weather?

Answer: Microwave

314. LiDAR is extensively used for:

Answer: Terrain elevation mapping

315. Passive sensors record:

Answer: Naturally reflected or emitted energy

316. Radar images are particularly useful for:

Answer: Flood and disaster mapping

317. Active remote sensing is advantageous because it:

Answer: Works at night

318. Microwave sensors are suitable for studying:

Answer: Surface roughness and moisture

319. Which sensor type measures time delay of returned signal?

Answer: Radar

320. Frequency and wavelength are related as:

Answer: Inversely proportional

321. Vegetation appears bright in:

Answer: Near infrared band

322. Which EM region is sensitive to surface temperature?

Answer: Thermal IR

323. Longer wavelengths generally have:

Answer: Lower energy

324. The visible region corresponds mainly to:

Answer: Human vision

325. Which EM region is absorbed most by atmospheric ozone?

Answer: Ultraviolet

326. Atmospheric windows are important because they:

Answer: Allow radiation to pass

327. Which band is useful for soil moisture studies?

Answer: Microwave

328. EM radiation travels at:

Answer: Speed of light

329. The spectral signature refers to:

Answer: Unique reflectance pattern of objects

330. A satellite with 1 m spatial resolution can detect:

Answer: Small objects clearly

331. Increasing radiometric resolution improves:

Answer: Brightness discrimination

332. Temporal resolution is critical in:

Answer: Disaster monitoring

333. A sensor with more spectral bands has higher:

Answer: Spectral resolution

334. Pixel is the smallest unit of:

Answer: Raster data

335. Hyperspectral data is useful for:

Answer: Detailed material identification

336. Trade-off often exists between:

Answer: Spatial and temporal resolution

337. Which resolution defines revisit frequency?

Answer: Temporal

338. Coarse resolution images are suitable for:

Answer: Global studies

339. Radiometric resolution is expressed in:

Answer: Bits

340. Scattering causes:

Answer: Image distortion

341. Clouds mainly affect:

Answer: Optical data

342. A geostationary satellite appears:

Answer: Fixed relative to Earth

343. IRS, Landsat, and Sentinel are examples of:

Answer: Earth observation satellites

344. GIS stores spatial data in:

Answer: Raster and vector formats

345. Large-scale maps are best for:

Answer: City-level planning

346. Map projection distortion occurs because Earth is:

Answer: Spherical

347. Spatial data answers questions related to:

Answer: Where

348. Digital cartography mainly uses:

Answer: Computers and GIS

349. Remote sensing + GIS is widely used in:

Answer: All of the above